

1 The Knapsack Problem

A hiker must select from among n items those items that will give him maximum utility. However, the hiker has only a knapsack in which to carry the items. The knapsack has a weight capacity of b lbs. Let c_i be the utility of item i , and let a_i be the weight of item i . The knapsack problem is to choose those items that will fit in the knapsack and at the same time maximize utility.

Variables. We define n binary variables x_i , $i = 1, \dots, n$ as

$$x_i = \begin{cases} 1, & \text{if item } i \text{ is selected} \\ 0, & \text{otherwise.} \end{cases}$$

Model.

$$\begin{aligned} \max \quad & \sum_{i=1}^n c_i x_i \\ \text{s.t.} \quad & \sum_{i=1}^n a_i x_i \leq b \\ & x_i \in \{0, 1\}, \quad i = 1, \dots, n. \end{aligned}$$

$$A = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

2 The Set-Covering Problem

Consider a set $U = \{1, 2, 3, 4, 5\}$ and a collection of its subsets $S_1 = \{1, 2, 3\}$, $S_2 = \{2, 4\}$, $S_3 = \{3, 4\}$ and $S_4 = \{4, 5\}$. The set-covering problem aims to find the minimum number of subsets to cover the set U .

More generally, let $U = \{1, \dots, m\}$ and let $S_1, \dots, S_n$ be n subsets of U . Let $A = (a_{ij})$ be the coefficient matrix such that $a_{ij} = 1$ if $i \in S_j$ and 0 otherwise. Also, let $c_j, j = 1, \dots, n$ be the costs of choosing set S_j in the selection. The goal of the problem is to choose a least-cost selection of subsets to cover all numbers in U .

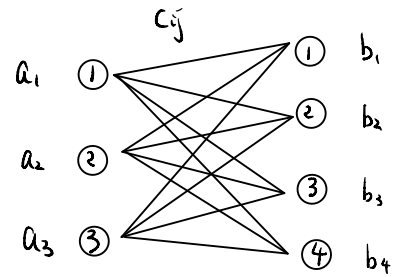
Variables. For each subset S_j , we define a binary variable x_j as

$$x_j = \begin{cases} 1, & \text{if set } S_j \text{ is selected} \\ 0, & \text{otherwise.} \end{cases}$$

Model.

$$\begin{aligned} \min \quad & \sum_{j=1}^n c_j x_j \\ \text{s.t.} \quad & \sum_{j=1}^n a_{ij} x_j \geq 1, \quad i = 1, \dots, m \\ & x_j \in \{0, 1\}, \quad j = 1, \dots, n. \end{aligned}$$

Obviously, the inequality constraints in the model guarantees that each number in U is covered by at least one selected subset.



3 The Facility Location Problem

There are m potential sites to construct new facilities, each of which is capable of producing a certain product. If constructed, each facility would serve as a source that would help supply the product to n destinations in order to satisfy the demands of the customers. If source i is constructed, then a one-time fixed charge of f_i is incurred, and it can produce at most a_i units of the product. Let b_j be the quantity demanded at destination j , and let c_{ij} be the unit production/shipping cost from source i to destination j . The goal of the problem is to minimize the total costs (fixed charges and production/shipping costs).

Variables. We need two types of variables — binary variables y_i and continuous variables x_{ij} .

$$y_i = \begin{cases} 1, & \text{if facility } i \text{ is constructed} \\ 0, & \text{otherwise,} \end{cases}$$

$$x_{ij} = \text{units shipped from source } i \text{ to destination } j.$$

Model.

$$\begin{aligned} \min \quad & \sum_{i=1}^m \left(f_i y_i + \sum_{j=1}^n c_{ij} x_{ij} \right) \\ \text{s.t.} \quad & \sum_{i=1}^m x_{ij} = b_j, \quad j = 1, \dots, n \\ & \sum_{j=1}^n x_{ij} \leq a_i y_i, \quad i = 1, \dots, m \\ & x_{ij} \geq 0, \quad i = 1, \dots, m, \quad j = 1, \dots, n \\ & y_i \in \{0, 1\}, \quad i = 1, \dots, m. \end{aligned}$$

The model is a variant of the transportation problem. The first set of constraints is demand constraints. For the second set of constraints, when $y_i = 1$, the constraint becomes a supply constraint; when $y_i = 0$, then x_{ij} is forced to be 0 for each $j = 1, \dots, n$.

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4 The Traveling Salesman Problem (TSP)

Starting from his home city, a salesman must visit each of n cities on a given list exactly once and return to his home city. The time taken to travel from city i to city j is c_{ij} . Find the order in which he should make his tour so as to finish as quickly as possible.

The TSP is a classical combinatorial problem with many applications. We introduce two mathematical formulations of the problem.

4.1 The Dantzig–Fulkerson–Johnson (DFJ) formulation

Variables. We define $n(n-1)$ binary variables x_{ij} , ($i, j = 1, \dots, n$, $i \neq j$) as

$$x_{ij} = \begin{cases} 1, & \text{if the salesman travels directly from city } i \text{ to city } j \\ 0, & \text{otherwise.} \end{cases}$$

Model.

$$\begin{aligned} \min \quad & \sum_{i=1}^n \sum_{\substack{j=1 \\ j \neq i}}^n c_{ij} x_{ij} \\ \text{s.t.} \quad & \sum_{\substack{j=1 \\ j \neq i}}^n x_{ij} = 1, \quad i = 1, \dots, n \\ & \sum_{\substack{i=1 \\ i \neq j}}^n x_{ij} = 1, \quad j = 1, \dots, n \\ & \sum_{i \in S} \sum_{j \in S} x_{ij} \leq |S| - 1, \quad S \subseteq \{1, \dots, n\}, \quad 2 \leq |S| \leq n-1 \\ & x_{ij} \in \{0, 1\}, \quad i, j = 1, \dots, n, \quad i \neq j. \end{aligned}$$

The first set of constraints requires the salesman to leave each city exactly once, while the second set of constraints requires the salesman to arrive at each city exactly once. The third set of constraints are called *subtour elimination constraints*. Their purpose is to ensure that a single tour results rather than a number of disjoint subtours. The DFJ subtour elimination constraints require that for each subset of cities, no small cycles (i.e. subtours) can be formed.

4.2 The Miller–Tucker–Zemlin (MTZ) formulation

Variables. We define $n(n-1)$ binary variables x_{ij} , ($i, j = 1, \dots, n$, $i \neq j$) the same as in the DFJ formulation. In addition, we introduce $(n-1)$ auxiliary variables u_i , ($i = 2, \dots, n$).

Model.

$$\begin{aligned}
\min \quad & \sum_{i=1}^n \sum_{\substack{j=1 \\ j \neq i}}^n c_{ij} x_{ij} \\
\text{s.t.} \quad & \sum_{\substack{j=1 \\ j \neq i}}^n x_{ij} = 1, \quad i = 1, \dots, n \\
& \sum_{\substack{i=1 \\ i \neq j}}^n x_{ij} = 1, \quad j = 1, \dots, n \\
& u_i - u_j + nx_{ij} \leq n-1, \quad i, j = 2, \dots, n, \quad i \neq j \\
& x_{ij} \in \{0, 1\}, \quad i, j = 1, \dots, n, \quad i \neq j.
\end{aligned}$$

The MTZ model differs from the DFJ model in the subtour elimination constraints. The MTZ subtour elimination constraints ensure that every tour contains city 1. For example, if cities 3, 4, and 5 form a subtour, then $x_{34} = x_{45} = x_{53} = 1$. The MTZ subtour elimination constraints contain the following inequalities:

$$\begin{aligned}
u_3 - u_4 + nx_{34} &\leq n-1 \\
u_4 - u_5 + nx_{45} &\leq n-1 \\
u_5 - u_3 + nx_{53} &\leq n-1.
\end{aligned}$$

Adding the three inequalities together, we have

$$n(x_{34} + x_{45} + x_{53}) \leq 3(n-1),$$

which is violated when $x_{34} = x_{45} = x_{53} = 1$.